

How Much Should You Feed Your Puppy?



Calories in food provide essential energy, for you as well as your pet. But how much should you be feeding your pet and how does the size and weight of your pet influence the amount of food they need?

- The adult weight of your dog is determined by the number of calories you feed your puppy.

The function $w = \left(\frac{k}{130}\right)^{\frac{4}{3}}$ relates the number of kilocalories per day, k , to the adult weight of the dog (in kilograms), w .

input: kcal
output: weight

- If you feed your puppy 800 kcal a day, will be their predicted adult weight?

$$w = \left(\frac{800}{130}\right)^{\frac{4}{3}} = 11.277 \text{ kg}$$

- Complete the table to predict the adult weight of your pet with each of the different diet plans. Round to the nearest tenth.

k (kcal a day)	600	800	1000
w (weight in kg)	7.7 kg	11.3 kg	15.2 kg

- If you want your dog to weigh 15.2 kg, how many calories should you feed him a day?

1000 kcal a day

- The ideal adult weight for a golden retriever is about 30 kg. How many calories a day should you feed a golden retriever to achieve this ideal weight? Show how you got your answer.

$$(30)^{\frac{3}{4}} = \left(\left(\frac{k}{130}\right)^{\frac{4}{3}}\right)^{\frac{3}{4}} \quad 130(30^{\frac{3}{4}}) = \left(\frac{k}{130}\right)^{\frac{4}{3}} 130$$

$$k = 130 (30)^{\frac{3}{4}} = 1666.4 \text{ kcal}$$

- The ideal adult weight for a Schnauzer is about 9 kg. How many calories a day should you feed a Schnauzer to achieve this ideal weight? Show how you got your answer.

$$(9)^{\frac{3}{4}} = \left(\left(\frac{k}{130}\right)^{\frac{4}{3}}\right)^{\frac{3}{4}} \quad 130(9^{\frac{3}{4}}) = 675.5 \text{ kcal}$$

- Write an equation that outputs the number of calories you should feed a dog whose ideal weight is w kg.

$$k = 130 w^{\frac{3}{4}}$$

inverse function **input: weight** **output: kcal**

- Explain how you could verify that your equation in question 4 gives you the correct number of calories.

Take the # of calories and plug it into the original equation to see if it gets you the desired weight

Lesson 1.9 – Inverse Functions

QuickNotes

Inverse functions "undo" the original function.

$$f^{-1}(f(c)) = c \quad f(f^{-1}(c)) = c$$

To find an inverse equation of $f(x)$

1) Replace $f(x)$ with y

2) Solve for the other variable.
(dependent variable of f^{-1})

3) Rewrite w/ inverse notation.

Check Your Understanding

1. Based on the table below, what is $g^{-1}(12)$?

x	4	8	12	16
$g(x)$	9	-2	7	12

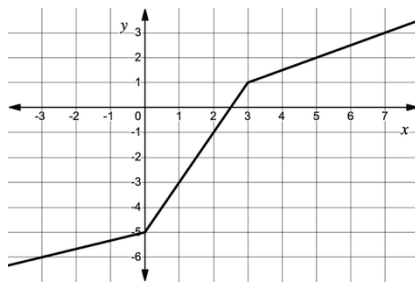
$$g^{-1}(12) = 16$$

since $g(16) = 12$

2. The graph of $y = h(x)$ is shown to the right. Find $h^{-1}(2)$.

$$h^{-1}(2) = 5$$

since $h(5) = 2$



3. To convert from degrees Fahrenheit to degrees Celsius, we can use the equation

$$C = \frac{5}{9}(F - 32).$$

Write the inverse of this function and what it represents.

$$F = \frac{9}{5}C + 32$$

It is the equation for converting from Celsius to Fahrenheit

4. The function $f(x)$ gives the distance in miles that an ambulance can travel within x minutes of receiving an emergency phone call. Interpret the meaning of $f^{-1}(26) = 20$ in context.

20 minutes have passed
by the time the ambulance has traveled 26 miles.

5. Let $g(x) = \frac{1}{x+2}$.

a. Find g^{-1} .

$$y = \frac{1}{x+2}$$

$$\frac{1}{y} = x+2$$

$$\frac{1}{y} - 2 = x$$

$$g^{-1}(y) = \frac{1}{y} - 2$$

b. Use compositions to verify that g and g^{-1} are in fact inverses.

$$\text{Need that } g^{-1}(g(x)) = x$$

$$\frac{1}{\left(\frac{1}{x+2}\right)} - 2 = x+2 - 2 = x \quad \checkmark$$